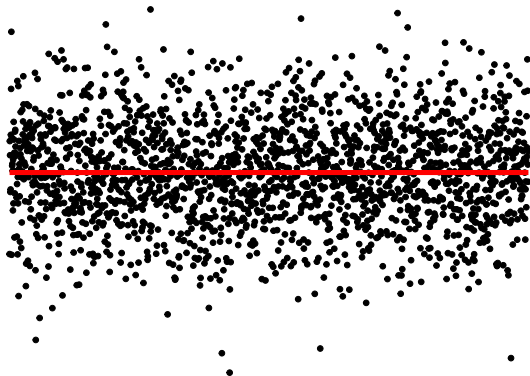


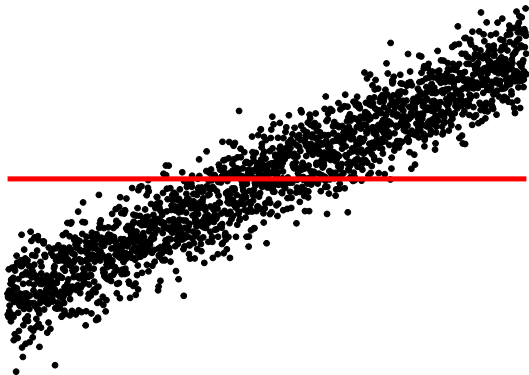


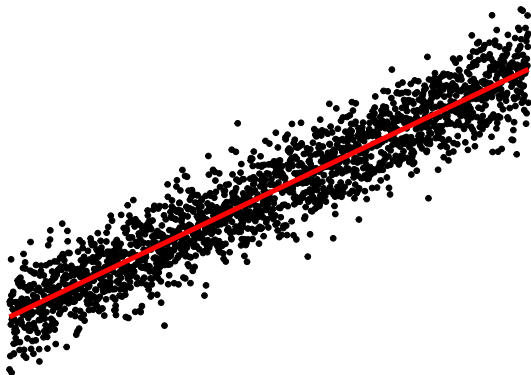
An introduction to the fundamentals of changepoint analysis

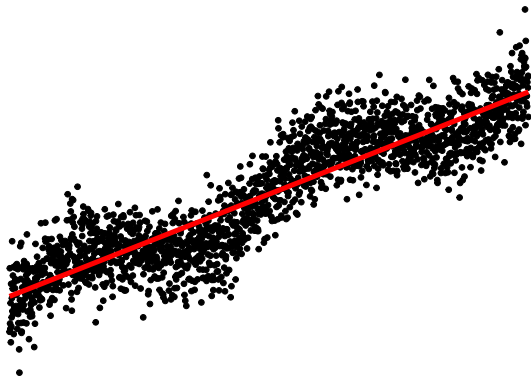
Rebecca Killick(r.killick@lancs.ac.uk)
NTSL Porquerolles, 2026

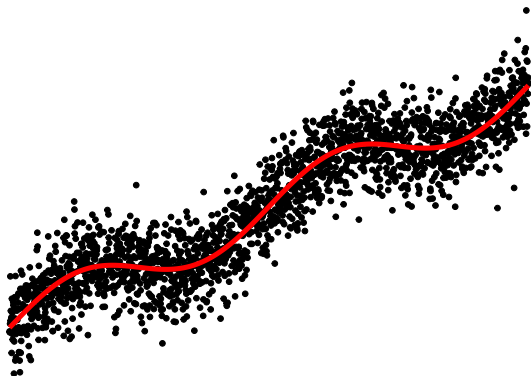




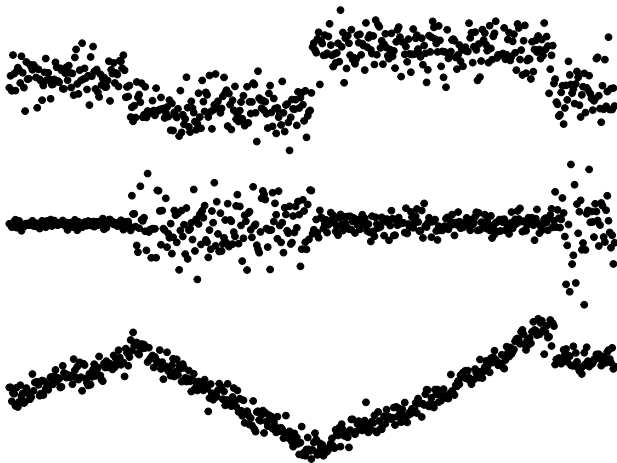








Model development



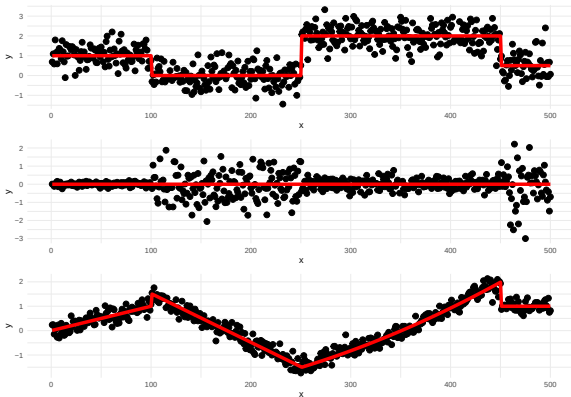
- What are changepoints?
- How do we fit changepoint models?
- Some key considerations
- Extensions

What are changepoints?



For data $y_1, \dots, y_n$, if a changepoint exists at τ , then $y_1, \dots, y_\tau$ differ from $y_{\tau+1}, \dots, y_n$ in some way.

There are many different types of change.

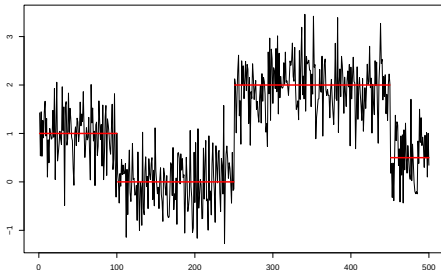




Assume we have time-series data where

$$Y_t | \theta_t \sim N(\theta_t, 1),$$

but where the means, θ_t , are piecewise constant through time.



We want to infer the number and position of the points at which the mean changes. One approach:

Likelihood Ratio Test

To detect a single changepoint we can use the likelihood ratio test statistic:

$$LR = \max_{\tau} \{ \ell(y_{1:\tau}) + \ell(y_{\tau+1:n}) - \ell(y_{1:n}) \}.$$

We infer a changepoint if $LR > \beta$ for some (suitably chosen) β . If we infer a changepoint its position is estimated as

$$\tau = \arg \max \{ \ell(y_{1:\tau}) + \ell(y_{\tau+1:n}) - \ell(y_{1:n}) \}.$$



Define m to be the number of changepoints, with positions $\tau = (\tau_0, \tau_1, \dots, \tau_{m+1})$ where $\tau_0 = 0$ and $\tau_{m+1} = n$.

- Multiple change hypothesis test?
- Binary segmentation
- Multiscale windowing

All the above have advantages and disadvantages but are inherently approximations as they conditionally detect changepoints.

Recall the likelihood ratio test decision for a change:

$$\max_{\tau} \{ \ell(y_{1:\tau}) + \ell(y_{\tau+1:n}) - \ell(y_{1:n}) \} > \beta.$$

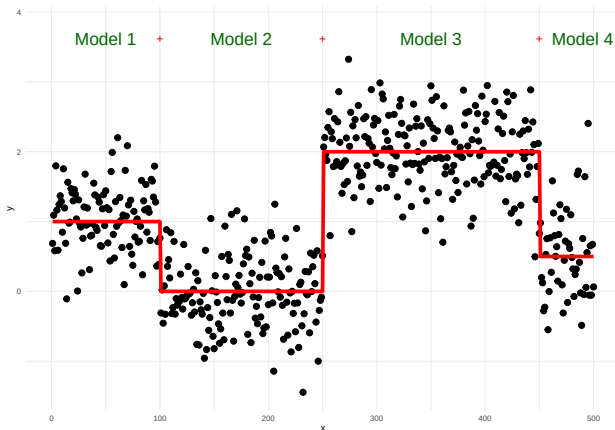
Then one application of the Likelihood ratio test can be viewed as

$$\min_{m \in \{0,1\}, \tau} \left\{ \sum_{i=1}^{m+1} \left[-\ell(y_{\tau_{i-1}:\tau_i}) \right] + \beta m \right\}$$

Repeated application is thus aiming to minimise

$$\min_{m, \tau} \left\{ \sum_{i=1}^{m+1} \left[-\ell(y_{\tau_{i-1}:\tau_i}) \right] + \beta m \right\}$$

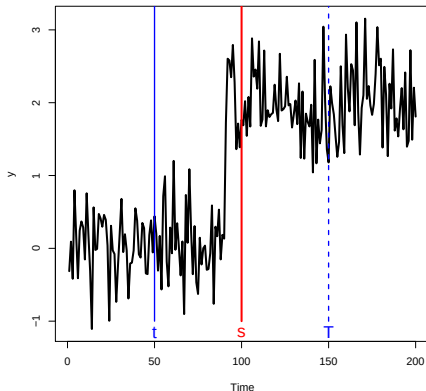
Problem



- How many changes?
- Where are the changes? 2^{n-1} possible solutions!



- Dynamic programming allows us to only worry about the location of the *last* change.
- Pruning means that as we go through the data we are smart about which locations are potential last change locations.



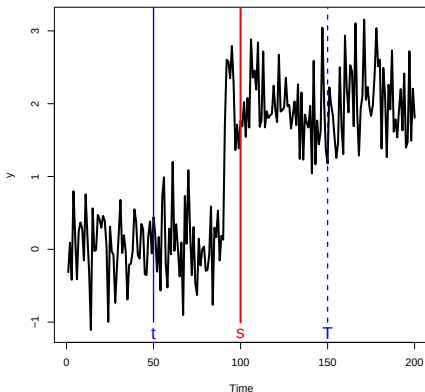


Let $0 < t < s < T$, if

$$F(t) + \mathcal{C}(y_{(t+1):s}) < F(s)$$

then at any future time $T > s$, t can never be the optimal last changepoint prior to T .

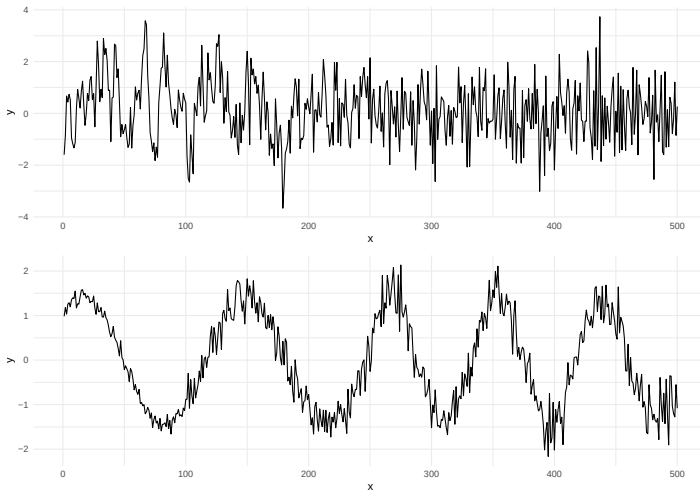
We can prove that, under certain regularity conditions, the expected computational complexity will be $\mathcal{O}(n)$.



- PDPA: looks at the optimizer as a function of the (single) parameter
- FPOP: combines PELT and PDRA for single parameter settings
- use computational geometry (convex hull) to compute the solution even faster
- factor in constraints (up-down-up, physics constrained)

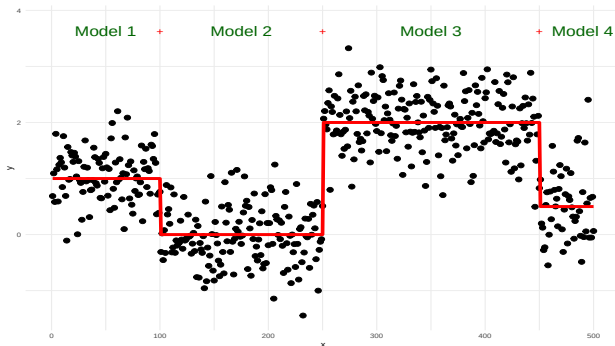
All of these approaches solve the optimization problem exactly.

More complicated changes



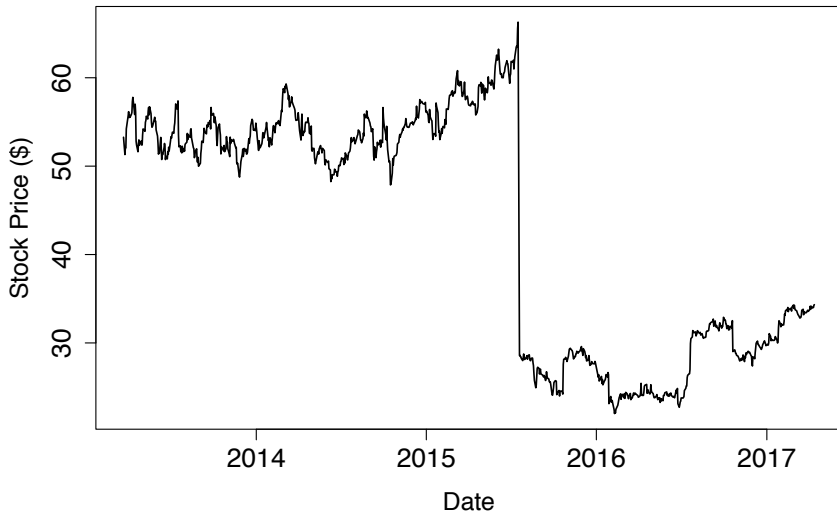


- Optimization
- Model development
- Theoretical properties
- Limits of detection
- Small sample and large sample in the same data



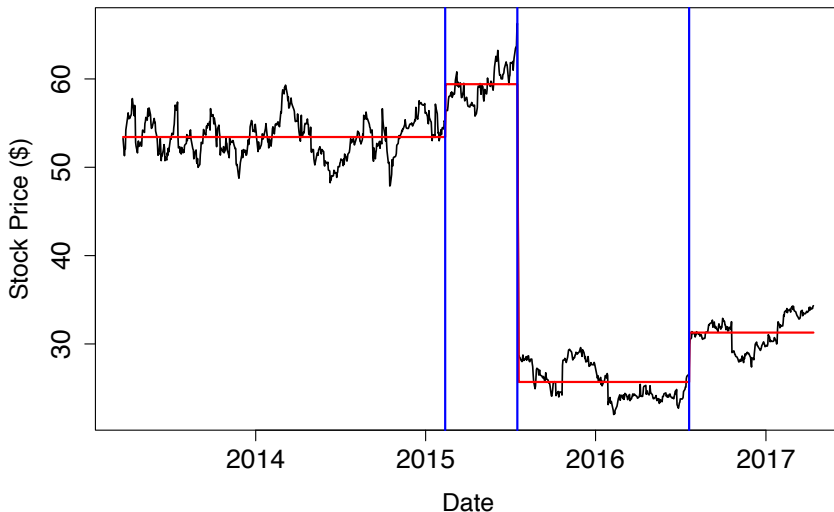


Changes are hard to identify with background correlation



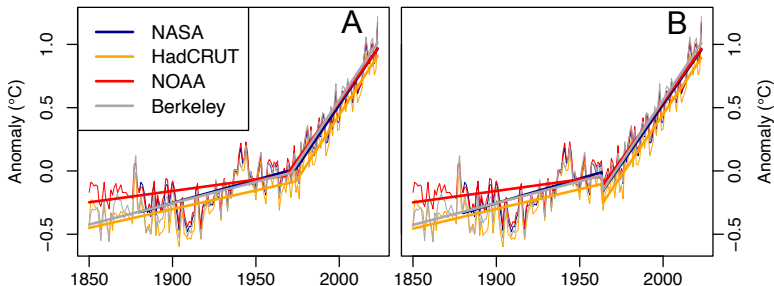


Change in mean with non-stationary errors





- Has there been a recent surge in global warming?
- Analysed several surface temperature time series estimates.

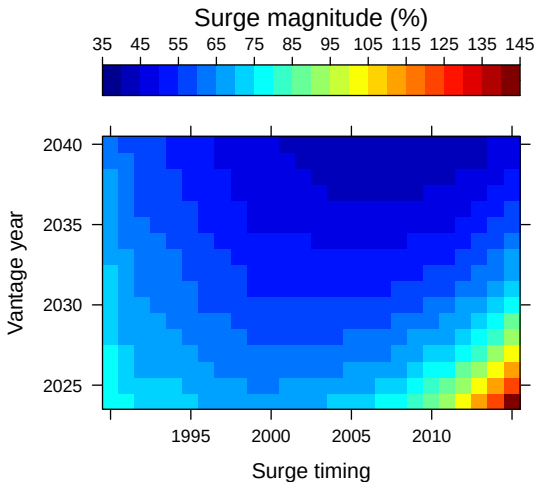


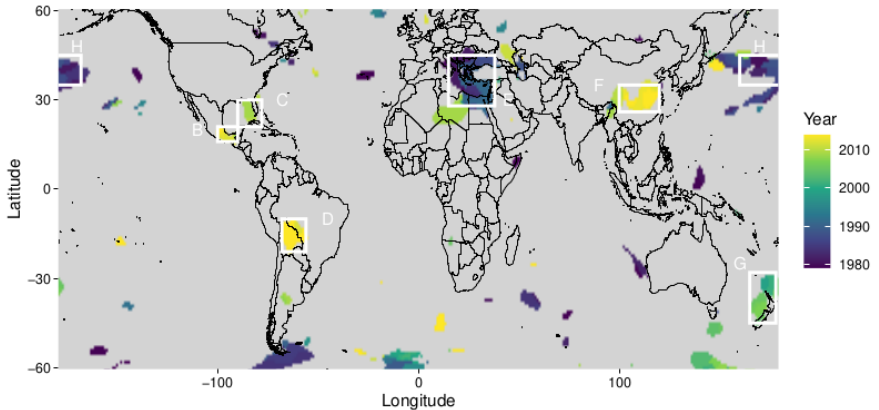
- Considered all possible models prevalent in the literature
- None show changes in the recent data

So what?



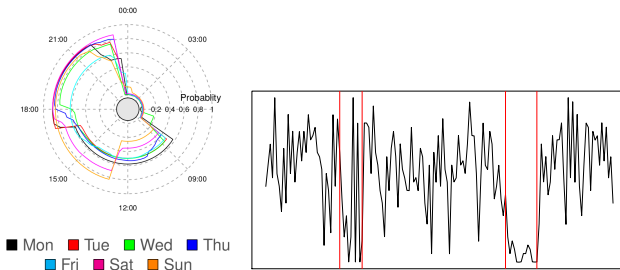
What would we need to see (in the statistically preferred model).







- Passively monitor patients at home
- Desire to have early intervention before crisis point is reached



- View data as circular, e.g., each day.
- Developed circular changepoint detection to detect changes in activity levels within a day
- Changes in the detected routine signal early intervention

What do we mean by a multivariate changepoint?

- Change occurs in all series at the same time?
- Change occurs in groups of series, are they known or unknown?
- Change is allowed to be in any combination at each change time?

What about model structures?

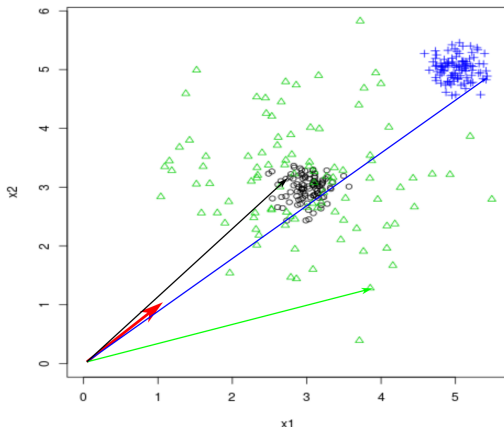
- Are the series correlated?
- If yes, does the correlation change too?
- Is the model form the same for each series?
- Are we assuming the same sampling for each series?

...

- Most approaches are using a projection to reduce dimension down to one.

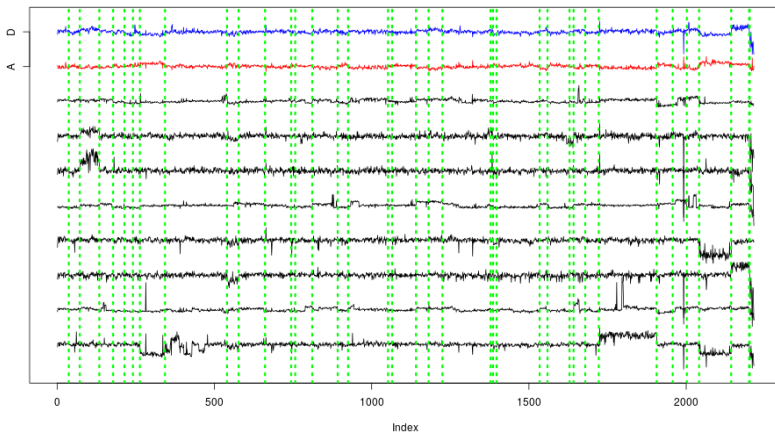


- Which genomic markers could be indications for disease?
- Mean and variance changes



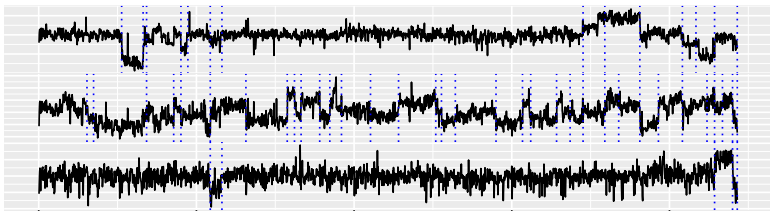


- Which genomic markers could be indications for disease?
- Use two transformations instead of one



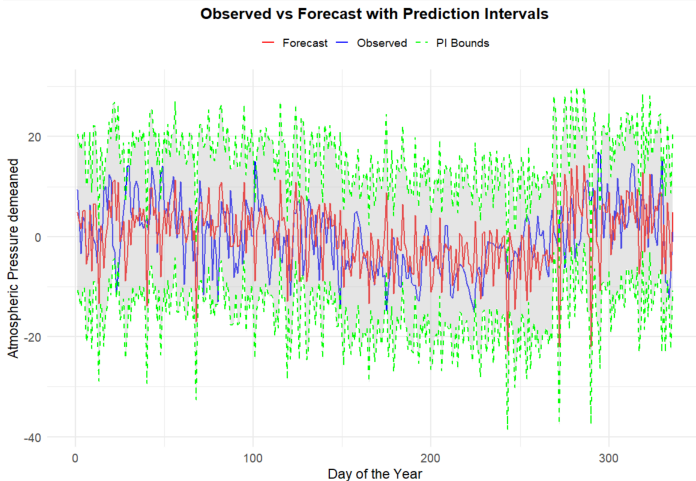


- Which genomic markers could be indications for disease?
- Only markers present in lots of patients are worth investigating.



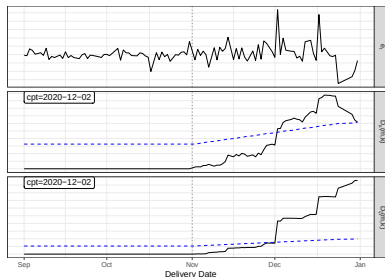
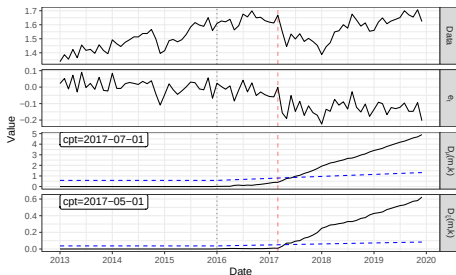
- Developed computationally efficient subset multivariate changepoint methodology.
- Only 10 changes are present in more than 60% of the patients.
- Over 50 detected by competitors.

Forecasting change



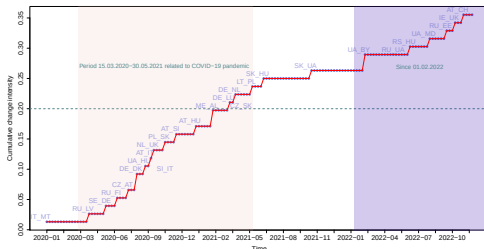
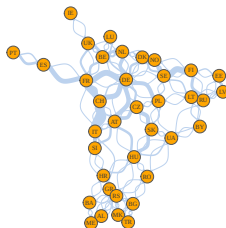


Monitor models for concept drift





Concept drift in time series on networks





- change in what? (models/model-free)
- loss function
- penalty / boundary for change
- search strategy for multiple changes



- Optimization
- Autocorrelation
- Multivariate data
- Model complexity
- Online changes

- Changepoints occur in most data
- Detecting and documenting them improves inference
- Multivariate, online, different data types mostly isn't a problem
- But there are still plenty of open problems in the field
- I enjoy working with messy real life data ...
- ... as often it sparks my next research challenge.

Preprints of all available at: www.lancs.ac.uk/~killick/pub.html

PELT: <https://doi.org/10.1080/01621459.2012.737745>

Model Choice: <https://doi.org/10.1175/JCLI-D-17-0863.1> &
<https://doi.org/10.1002/qre.2712>

LMvsCpts: <https://doi.org/10.1007/s11222-017-9731-0> &
<https://doi.org/10.1002/env.2568>

Multivariate: <https://doi.org/10.1038/s41598-019-46836-y> &
<https://doi.org/10.1007/s11222-020-09940-y>

Autocorrelation: <https://doi.org/10.1002/sta4.351>

Circular time: <http://doi.org/10.1111/rssc.12472>

Sensitivity: <https://doi.org/10.1080/10618600.2021.2000873>

Multivariate: <https://doi.org/10.1080/00401706.2023.2183261>

Autocorrelation: <https://doi.org/10.1007/s42952-022-00173-5>

Warming Surge: <https://doi.org/10.1038/s43247-024-01711-1>

Concept drift: <https://doi.org/10.1111/jtsa.12843>

Networks: <https://arxiv.org/abs/2312.16357>

Change in variance with nonstationary errors:

<https://doi.org/10.1002/env.2576>

Changes in second order structure within nonstationary series:

<https://doi.org/10.1214/13-EJS799>

Uncertainty of autocovariance changes:

<https://doi.org/10.1080/00401706.2014.902776>

Links or stars from: <https://www.github.com/rkillick>

Online: <https://github.com/grundy95/changepoint.forecast>

Multivariate:

Covariance:

https://github.com/s-ryan1/Covariance_RMT_simulations

Mean/Var: <https://github.com/grundy95/changepoint.geo>